

Introduction to the Classical Shadow Tomography

Mingqian Chen

December 18, 2023

Abstract

Accurately predicting target observables and understanding the properties of a quantum system are crucial in developing quantum algorithms and technologies. The concept of shadow tomography, originally introduced by Aaronson [2], allows for learning target observables without obtaining a complete quantum state description. Based on this idea, Huang et al. proposed the classical shadow protocol [6], which enables the prediction of many target observables with only a logarithmic number of measurements in the number of observables. This report is an introduction to the classical shadow tomography.

1 Introduction

Quantum state tomography is a method used to reconstruct a quantum state, providing a complete description through a series of measurements on identical copies of the state in various bases. Unlike a classical object that can be measured repeatedly, measurements of quantum states are destructive. Each measurement offers only partial information, as the quantum system's wave function collapses upon measurement, leading to probabilistic outcomes by Born's rule. Consequently, one must take many measurements on identical copies of an unknown state for a complete description. However, due to a curse of dimensionality, the number of parameters needed to fully reconstruct a state scales exponentially with the system size. Haah et al. showed that an order of $O(d^2/\epsilon^2)$ copies is sufficient to recover a description of a d -dimensional quantum state ρ [7] by taking trace norm. Efficient quantum tomography shows that $O(d/\epsilon)$ copies are sufficient to reconstruct a quantum state with high accuracy when taking the Frobenius norm [1], and the result is equivalent to the sample optimal protocol [7] when taking the trace norm. However, for a n qubit quantum system, $d = 2^n$ will still be infeasible.

Aaronson pointed out that many observables can be predicted simultaneously even without obtaining a complete description of a quantum state, and first proposed the shadow tomography procedure [2], which allows estimating many observables with only $\tilde{O}(\log^4 M \cdot \log D)$ state copies. This bound was later improved to $O(\log^2 M \cdot \log^2 D)$ by the gentle measurement proposed by Aaronson and Rothblum [4], where M is the number of measurements, and $D = 2^n$ is the dimension of the system. Though this procedure significantly reduced

the sample complexity, it requires large quantum memory to store the state copies and exponentially long quantum circuits to make the predictions.

1.1 Classical shadow

Based on the idea of Aaronson’s shadow tomography, Huang et al. proposed a more time-efficient procedure called classical shadow tomography and reduced the number of **copies to $\log(M)$ where M is the number of target functions to be predicted** [6]. Unlike the previous algorithms, the sample complexity is **independent of the system size**. The procedure of predicting quantum observables has two parts: the **data acquisition** process and the **prediction phase**. The data acquisition is to obtain the classical descriptions of the unknown state and a measurement channel from the state to its classical descriptions. Given an unknown n -qubit state ρ , **randomly rotate the state by acting on a unitary randomly sampled from a unitary ensemble**. By taking measurements in the computational basis, the states **collapse to n -qubit bit strings $|\hat{b}\rangle \in \{0,1\}^n$** . Then, by applying $U^\dagger$ of the corresponding unitaries on the measurement outcomes, we obtain the snapshots $U^\dagger |\hat{b}\rangle \langle \hat{b}| U$ of the underlying quantum state. The probability of measuring each outcome is calculated by Born’s rule, $\Pr[\hat{b} = b] = \langle b| U \rho U^\dagger |b\rangle$. Each snapshot is related to a specific property of the unknown quantum state and is stored in classical memory. A measurement channel can describe the mapping from the unknown state to the snapshot in expectation:

$$\mathbb{E}[U^\dagger |\hat{b}\rangle \langle \hat{b}| U] = \mathbb{E}_{U \sim \mathcal{U}} \sum_{b \in \{0,1\}^n} \langle b| U \rho U^\dagger |b\rangle U^\dagger |b\rangle \langle b| U = \mathcal{M}(\rho). \quad (1)$$

Note that **the unitary ensemble is assumed to be tomographically complete**, meaning a quantum state can be fully reconstructed based on measurement outcomes. A unitary ensemble of tomographically complete implies that, by applying different unitaries from the ensemble to a quantum state and taking measurements on the rotated states, one can acquire enough information to determine the initial quantum state uniquely. In this case, the measurement channel $\mathcal{M}$ is reversible, and the reversed map $\mathcal{M}^{-1}$ is called the reconstruction map, which reconstructs a classical shadow by its corresponding snapshot

$$\hat{\rho} = \mathcal{M}^{-1}(U^\dagger |b\rangle \langle b| U), \quad (2)$$

and taking the average of the classical shadow over unitaries and outcome distribution recovers the underlying quantum state

$$\mathbb{E}[\hat{\rho}] = \rho. \quad (3)$$

The prediction process uses the median of means method to avoid statistical fluctuations. Perform the previous steps N times, and one obtains an array of N independent classical shadows

$$\mathcal{S}(\rho; N) = \{\hat{\rho}_1, \hat{\rho}_1, \dots, \hat{\rho}_N\}. \quad (4)$$

Split the shadows into K groups

$$\hat{\rho}_{(k)} = \frac{1}{\lfloor N/K \rfloor} \sum_{i=(k-1)\lfloor N/K \rfloor + 1}^{k\lfloor N/K \rfloor} \hat{\rho}_i, \quad (5)$$

and predict each observable $O_1, \dots, O_M$ by taking the median of the expectations

$$\hat{o}_i(N, K) = \text{median}\{\text{tr}(O_i \hat{\rho}_{(1)}), \dots, \text{tr}(O_i \hat{\rho}_{(K)})\}. \quad (6)$$

Theorem 1 *The protocol of the classical shadow allows one to predict M target functions $\text{tr}(O_i \rho)$ within error ϵ by at least $O(\log(M) \max_i \|O_i\|_{\text{shadow}}^2 / \epsilon^2)$ measurements.*

Classical shadow sampled from different unitaries has different shadow norms. The shadow norm $\|O_i\|_{\text{shadow}}^2$ implies the sample complexity and variance in prediction. For random Clifford measurements, $\|O_i\|_{\text{shadow}}^2$ is equivalent to the Hilbert-Schmidt norm, and for random Pauli measurement, $\|O_i\|_{\text{shadow}}^2$ scales exponentially in the subsystem size.

1.1.1 Measurement Channel and Shadow Norm

It has been shown that the n -qubit Clifford group $\text{Cl}(2^n)$ is a unitary 3-design [8] [9] [10]. Random sampling of the Clifford group is equivalent to random sampling of the Haar unitary at the first 3 moments

$$\mathbb{E}_{U \sim \text{Cl}(2^n)} (UXU^\dagger)^{\otimes k} = \int_{U(d)} (UAU^\dagger)^{\otimes k} d\mu_{\text{Haar}}(U), \quad (7)$$

where A are $2^n \times 2^n$ matrices. Taking average over random Clifford circuits, the expectation of linear operators yields

$$\mathbb{E}_{U \sim \text{Cl}(2^n)} U^\dagger |x\rangle \langle x| U \langle x| UAU^\dagger |x\rangle = \frac{A + \text{tr}(A)\mathbb{I}}{(2^n + 1)2^n} = \frac{1}{2^n} \mathcal{D}_{1/(2^n+1)}(A) \quad (8)$$

for $A \in \mathbb{H}_{2^n}$ where $\mathbb{H}_{2^n}$ is the $2^n \times 2^n$ Hermitian matrix space. The depolarizing channel is invertible

$$\mathcal{D}_{1/(2^n+1)}^{-1}(A) = (2^n + 1)A - \text{tr}(A)\mathbb{I}. \quad (9)$$

The expectation of quadratic operators yields

$$\mathbb{E}_{U \sim \text{Cl}(2^n)} U^\dagger |x\rangle \langle x| U \langle x| UB_0U^\dagger |x\rangle U \langle x| UC_0U^\dagger |x\rangle = \frac{\text{tr}(B_0C_0)\mathbb{I} + B_0C_0 + C_0B_0}{(2^n + 2)(2^n + 1)2^n} \quad (10)$$

for $B_0, C_0 \in \mathbb{H}_{2^n}$ traceless.

The variance of predicting a linear observable O , $\text{Var}[\hat{o}] = \mathbb{E}[(\hat{o} - [\hat{o}])^2] \leq \left\| O - \frac{\text{tr}(O)}{2^n} \mathbb{I} \right\|_{\text{shadow}}^2$ is bounded by the shadow norm, and the shadow norm depends on the measurement scheme

$$\|O\|_{\text{shadow}} = \max_{\sigma: \text{state}} (\mathbb{E}_{U \sim \mathcal{U}} \sum_{b \in \{0,1\}^n} \langle b| U \sigma U^\dagger |b\rangle \langle b| U \mathcal{M}^{-1}(O) U^\dagger |b\rangle^2)^{1/2}. \quad (11)$$

Random Clifford Measurements In this case, the unitary ensemble is the n -qubit Clifford group $\text{Cl}(2^n)$. Combining eq. (1) and (8), one can obtain the mapping channel

$$\mathcal{M}(\rho) = \sum_{b \in \{0,1\}^n} \mathbb{E}_{U \sim \text{Cl}(2^n)} \langle b | U \rho U^\dagger | b \rangle U^\dagger | b \rangle \langle b | U = \mathcal{D}_{1/(2^n+1)}(\rho), \quad (12)$$

and the classical shadow reconstructed by $\mathcal{M}^{-1}$ is

$$\hat{\rho} = \mathcal{M}^{-1}(U^\dagger |\hat{b}\rangle \langle \hat{b}| U) = (2^n + 1) U^\dagger |\hat{b}\rangle \langle \hat{b}| U - \mathbb{I}. \quad (13)$$

Since classical shadows have unit trace, the shadow norm depends only on the traceless part of the observable. Combining the definition of the shadow norm eq. (11) and eq. (10)

$$\|O_0\|_{\text{shadow}}^2 = \max_{\sigma: \text{state}} \text{tr}(\sigma \sum_{b \in \{0,1\}^n} \mathbb{E}_{U \sim \text{Cl}(2^n)} U^\dagger |b\rangle \langle b| U \langle b| U (2^n + 1) O_0 U^\dagger |b\rangle^2). \quad (14)$$

The shadow norm is bounded by the Hilbert-Schmidt norm for any traceless $O_0 \in \mathbb{H}_{2^n}$ by further simplification

$$\text{tr}(O_0^2) \leq \|O_0\|_{\text{shadow}}^2 \leq 3\text{tr}(O_0^2) \quad (15)$$

In this case, the shadow norm is bounded by the trace of O_0^2 , and is efficient in estimating global observables such as quantum fidelity.

Random Pauli Measurements Unlike the previous case the Clifford gate is applied globally, in this case, each qubit is rotated by randomly sampled single-qubit Clifford gates $U_1 \otimes \dots \otimes U_n$. This process is equivalent to taking random Pauli measurements on each qubit. By the nature of the measurement primitives, the mapping channel of eq. (1) can be decomposed into a tensor product of single qubit operations $X_1 \otimes \dots \otimes X_n \in \mathbb{H}_2^{\otimes n}$, thus we apply eq. (8) on each qubit

$$\mathcal{M}(X_1 \otimes \dots \otimes X_n) = \bigotimes_{j=1}^n \left(\sum_{b_j \in \{0,1\}} \mathbb{E}_{U_j \sim \text{Cl}(2)} U_j^\dagger |b_j\rangle \langle b_j| U_j \langle b_j| U_j X_j U_j^\dagger |b_j\rangle \right). \quad (16)$$

The classical shadow is

$$\hat{\rho} = \mathcal{M}^{-1}(U^\dagger |\hat{b}\rangle \langle \hat{b}| U) = \bigotimes_{j=1}^n (3U_j^\dagger |\hat{b}_j\rangle \langle \hat{b}_j| U_j - \mathbb{I}). \quad (17)$$

For a k -local observable, the shadow norm defined in eq. 11 becomes

$$\begin{aligned} \left\| \tilde{O} \otimes \mathbb{I}^{\otimes(n-k)} \right\|_{\text{shadow}}^2 &= \max_{\sigma: \text{state}} \mathbb{E}_{U \sim \text{Cl}(2)^{\otimes n}} \sum_{b \in \{0,1\}^n} \langle b | U \sigma U^\dagger | b \rangle \langle b | U \mathcal{M}^{-1}(\tilde{O} \otimes \mathbb{I}^{\otimes(n-k)}) U^\dagger | b \rangle^2 \\ &= \max_{\sigma: \text{state}} \mathbb{E}_{U \sim \text{Cl}(2)^{\otimes k}} \sum_{b \in \{0,1\}^k} \langle b | U \text{tr}_{k+1, \dots, n}(\sigma) U^\dagger | b \rangle \langle b | U \tilde{\mathcal{M}}^{-1}(\tilde{O}) U^\dagger | b \rangle^2. \end{aligned} \quad (18)$$

For a single k -local observable, the shadow norm scales exponentially with locality

$$\|O\|_{\text{shadow}}^2 = 3^k. \quad (19)$$

For an in-depth computation and derivation, please refer to the paper by Huang et al [6].

1.1.2 Numerical Results

Random Clifford Measurements In Figure 1, Huang et al implemented neural network quantum state tomography (NNQST) [11] and classical shadow for GHZ states, and compared their performance in the prediction of quantum fidelities. The plot on the left is the number of experiments needed to identify an n -qubit GHZ state with 0.99 fidelity. The number of measurements required by NNQST scales linearly with the system size, whereas it remains constant for the classical shadow. This result confirms the theorem that the size of the classical shadow based on random Clifford measurements is independent of the system size when estimating global observables. The plot on the right compares the ability of both approaches to detect errors in state preparation. The fidelity of both noisy state and pure state are compared. The pure GHZ state is $|\psi_{\text{GHZ}}^{\pm}(n)\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes n} \pm |1\rangle^{\otimes n})$, and a phase error occurs in the state preparation of the noisy GHZ state with a probability of p , $\rho_p = (1-p)|\psi_{\text{GHZ}}^+(n)\rangle\langle\psi_{\text{GHZ}}^+(n)| + p|\psi_{\text{GHZ}}^-(n)\rangle\langle\psi_{\text{GHZ}}^-(n)|$. When $p = 1$, the noisy state becomes orthogonal to the pure state, and classical shadow estimates 0 fidelity while NNQST constantly overestimates the fidelity. The result shows that the classical shadow not only requires fewer measurements but also yields better accuracy.

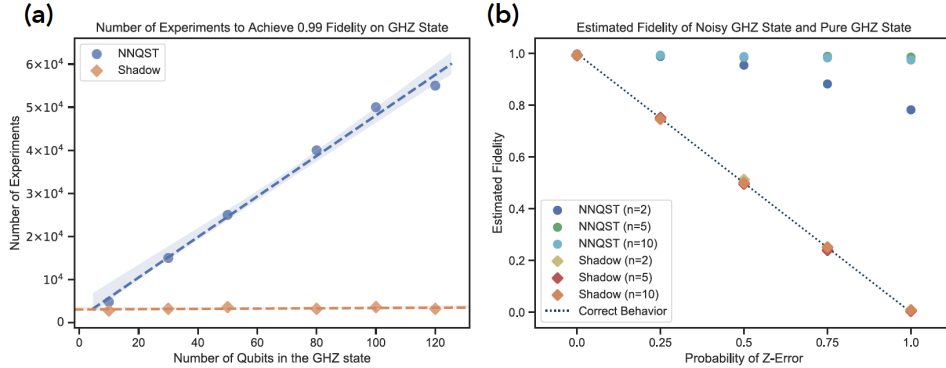


Figure 1: Comparison of classical shadow and NNQST in predicting quantum fidelities [6].

Random Pauli Measurements The plot on the left of Figure 2 presents the numerical results obtained by Huang et al., comparing the classical shadow method with the Brydges et al. protocol [12] for estimating second-order Rényi entropies across all subsystems up to size 2 within the approximate ground state of a 1D disordered Heisenberg spin chain. Both approaches involve performing randomized measurements on each qubit and taking computational basis measurements. The classical shadow, obtained with 2500 measurements, outperforms the Brydges et al. protocol by achieving a maximum error of 0.052. In comparison, the Brydges et al. protocol yields a maximum prediction error of 0.24 [6]. The plot on the right of Figure 2 demonstrates that fewer measurements are needed with the classical shadow approach to predict the second-order Rényi entanglement entropy of GHZ states in comparison to the Brydges et al. protocol.

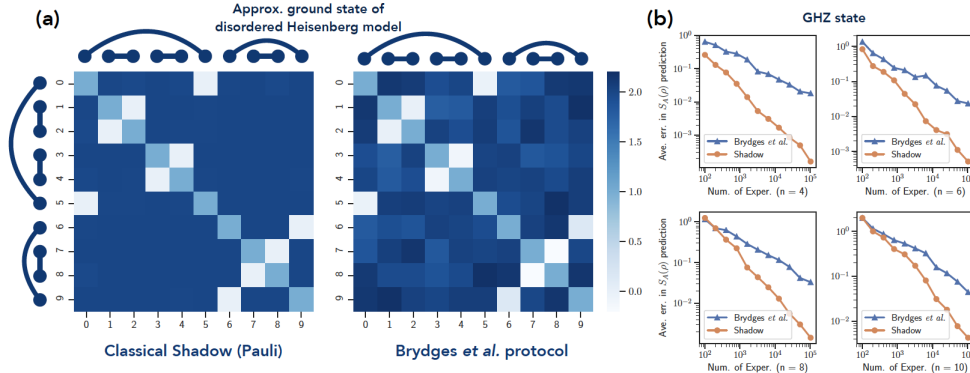


Figure 2: Comparison of classical shadow and Brydeges et al. protocol in estimating Rényi entanglement entropy [6].

References

- [1] Wright, J. (2015). Efficient quantum tomography. ArXiv. /abs/1508.01907
- [2] Aaronson, S. (2017). Shadow Tomography of Quantum States. ArXiv. <https://doi.org/10.48550/arXiv.1711.01053>
- [3] Brandão, F. G., Kalev, A., Li, T., Lin, C. Y., Svore, K. M., & Wu, X. (2017). Quantum SDP Solvers: Large Speed-ups, Optimality, and Applications to Quantum Learning. ArXiv. /abs/1710.02581
- [4] Aaronson, S., & Rothblum, G. N. (2019). Gentle Measurement of Quantum States and Differential Privacy. ArXiv. /abs/1904.08747
- [5] Huang, H., Kueng, R., & Preskill, J. (2021). Efficient estimation of Pauli observables by derandomization. ArXiv. <https://doi.org/10.1103/PhysRevLett.127.030503>
- [6] Huang, H., Kueng, R., & Preskill, J. (2020). Predicting Many Properties of a Quantum System from Very Few Measurements. ArXiv. <https://doi.org/10.1038/s41567-020-0932-7>
- [7] Haah, J., Harrow, A. W., Ji, Z., Wu, X., & Yu, N. (2015). Sample-optimal tomography of quantum states. ArXiv. <https://doi.org/10.1109/TIT.2017.2719044>
- [8] Zhu, H. (2015). Multiqubit Clifford groups are unitary 3-designs. ArXiv. <https://doi.org/10.1103/PhysRevA.96.062336>
- [9] Kueng, R., & Gross, D. (2015). Qubit stabilizer states are complex projective 3-designs. ArXiv. /abs/1510.02767
- [10] Webb, Z. (2015). The Clifford group forms a unitary 3-design. ArXiv. /abs/1510.02769
- [11] Renes, J. M., Scott, A. J., & Caves, C. M. (2003). Symmetric Informationally Complete Quantum Measurements. ArXiv. <https://doi.org/10.1063/1.1737053>

- [12] Brydges, T., Elben, A., Jurcevic, P., Vermersch, B., Maier, C., Lanyon, B. P., Zoller, P., Blatt, R., & Roos, C. F. (2018). Probing entanglement entropy via randomized measurements. ArXiv. <https://doi.org/10.1126/science.aau4963>
- [13] Xin, T., Lu, S., Cao, N., Anikeeva, G., Lu, D., Li, J., Long, G., & Zeng, B. (2018). Local-measurement-based quantum state tomography via neural networks. ArXiv. <https://doi.org/10.1038/s41534-019-0222-3>
- [14] Torlai, G., Mazzola, G., Carrasquilla, J., Troyer, M., Melko, R., & Carleo, G. (2017). Many-body quantum state tomography with neural networks. ArXiv. <https://doi.org/10.1038/s41567-018-0048-5>
- [15] Aharonov, D., Ambainis, A., Kempe, J., & Vazirani, U. (2000). Quantum Walks On Graphs. ArXiv. <https://doi.org/10.48550/arXiv.quant-ph/0012090>
- [16] Feynman, R. & Hibbs, A. Quantum mechanics and path integrals. (McGraw-Hill,1965), <https://cds.cern.ch/record/100771>
- [17] Huang, H., Kueng, R., & Preskill, J. (2021). Efficient estimation of Pauli observables by derandomization. ArXiv. <https://doi.org/10.1103/PhysRevLett.127.030503>